TEST REPORT

CheckCode:088382
Report No.:C202404125775-4E

Company Name: SIMCom Wireless Solutions Limited
Address: SIMCom Headquarters Building, Building 3, No.289 Linhong Road, Shanghai, China

The following sample information is provided by the client and responsible for its authenticity

Sample Name: LPWA Module
Type/Model: SIM7070G
Sample State: Normal

Date of Receipt: Jun.6, 2024 Test Period: Jun.6, 2024 – Jun.17, 2024

Test Request: As specified by client, to determine the Lead(Pb),Cadmium(Cd),Mercury(Hg), Hexavalent Chromium [Cr(Ⅵ)], Polybrominated biphenyls (PBBs), Polybrominated diphenyl ethers (PBDEs) , Bis(2-ethylhexyl) phthalate (DEHP) , Butyl benzyl phthalate (BBP), Dibutyl phthalate (DBP) , Diisobutyl phthalate (DIBP)content in the submitted sample(s).

Judge Standard: RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU.

Test Result: Please refer to following page(s).

Conclusion: Based on the performed test on submitted sample, the result(s) comply with the limits as set by RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU.

Edited by Su Huiying Reviewed by Wu Yanyan Approved by Wang Sheng

扫一扫 验真伪

Issue date:Jun.17,2024

The result(s) shown in this report refer(s) only to the sample(s) tested. This report is considered invalid without approved signature, special seal for inspection and testing and the seal on the perforation.This report cannot be reproduced except in full, without prior written approval of the Company.Any disagreements of the test report should be fed back to us within 15 workdays upon receiving the report.

GRG METROLOGY & TEST (WUXI) CO., LTD.
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Tel: 4006020999 Fax: +86-0510-68002628 Website: http://www.grgtest.com

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Test Result:

Test Methods:

Test Item(s)	Test Method	Measuring Instrument	MDL	Limit
Pb, Cd, Hg, Total Cr, Total Br	IEC 62321-3-1:2013	XRF	20mg/kg	/
Lead (Pb)	IEC 62321-5:2013	ICP-OES	2mg/kg	1000mg/kg
Cadmium (Cd)	IEC 62321-5:2013	ICP-OES	2mg/kg	100mg/kg
Mercury (Hg)	IEC 62321-4:2013 +AMD1:2017	ICP-OES	2mg/kg	1000mg/kg
Hexavalent Chromium [Cr(Ⅵ)]	IEC 62321-7-2:2017	UV-Vis	8mg/kg	1000mg/kg
	IEC 62321-7-1:2015	UV-Vis	0.10μg/cm^2	/
Polybrominated biphenyls (PBBs)	IEC 62321-6:2015	GC-MS	50mg/kg	1000mg/kg
Polybrominated diphenyl ethers (PBDEs)	IEC 62321-6:2015	GC-MS	50mg/kg	1000mg/kg
Bis(2-ethylhexyl) phthalate (DEHP)	IEC 62321-8:2017	GC-MS	50mg/kg	1000mg/kg
Butyl benzyl phthalate (BBP)	IEC 62321-8:2017	GC-MS	50mg/kg	1000mg/kg
Dibutyl phthalate (DBP)	IEC 62321-8:2017	GC-MS	50mg/kg	1000mg/kg
Diisobutyl phthalate (DIBP)	IEC 62321-8:2017	GC-MS	50mg/kg	1000mg/kg

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No.	Sample Description	Restricted Substances	Results of XRF(mg/kg)	Result of Wet Chemical Testing(mg/kg)	Conclusion on RoHS
1	White label with black/red lettering	Pb	BL	/	Comply
		Cd	BL	/	Comply
		Hg	BL	/	Comply
		Cr(Ⅵ)	BL	/	Comply
		PBBs	BL	/	Comply
		PBDEs		/	Comply
		DEHP	/	N.D.	Comply
		BBP	/	N.D.	Comply
		DBP	/	N.D.	Comply
		DIBP	/	N.D.	Comply
2	Silvery-grey metal shell	Pb	BL	/	Comply
		Cd	BL	/	Comply
		Hg	BL	/	Comply
		Cr(Ⅵ)	BL	/	Comply
		PBBs	/	/	/
		PBDEs		/	/
		DEHP	/	/	/
		BBP	/	/	/
		DBP	/	/	/
		DIBP	/	/	/
3	White/green body	Pb	BL	/	Comply
		Cd	BL	/	Comply
		Hg	BL	/	Comply
		Cr(Ⅵ)	BL	/	Comply
		PBBs	BL	/	Comply
		PBDEs		/	Comply
		DEHP	/	N.D.	Comply
		BBP	/	N.D.	Comply
		DBP	/	N.D.	Comply
		DIBP	/	N.D.	Comply

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No.	Sample Description	Restricted Substances	Results of XRF(mg/kg)	Result of Wet Chemical Testing(mg/kg)	Conclusion on RoHS
4	Brown body	Pb	BL	/	Comply
		Cd	BL	/	Comply
		Hg	BL	/	Comply
		Cr(Ⅵ)	BL	/	Comply
		PBBs	BL	/	Comply
		PBDEs		/	Comply
		DEHP	/	N.D.	Comply
		BBP	/	N.D.	Comply
		DBP	/	N.D.	Comply
		DIBP	/	N.D.	Comply
5	Black body	Pb	BL	/	Comply
		Cd	BL	/	Comply
		Hg	BL	/	Comply
		Cr(Ⅵ)	BL	/	Comply
		PBBs	BL	/	Comply
		PBDEs		/	Comply
		DEHP	/	N.D.	Comply
		BBP	/	N.D.	Comply
		DBP	/	N.D.	Comply
		DIBP	/	N.D.	Comply
6	Black body	Pb	BL	/	Comply
		Cd	BL	/	Comply
		Hg	BL	/	Comply
		Cr(Ⅵ)	BL	/	Comply
		PBBs	BL	/	Comply
		PBDEs		/	Comply
		DEHP	/	N.D.	Comply
		BBP	/	N.D.	Comply
		DBP	/	N.D.	Comply
		DIBP	/	N.D.	Comply

No.	Sample Description	Restricted Substances	Results of XRF(mg/kg)	Result of Wet Chemical Testing(mg/kg)	Conclusion on RoHS
7	Green PCB plate	Pb	BL	/	Comply
		Cd	BL	/	Comply
		Hg	BL	/	Comply
		Cr(Ⅵ)	BL	/	Comply
		PBBs	BL	/	Comply
		PBDEs		/	Comply
		DEHP	/	N.D.	Comply
		BBP	/	N.D.	Comply
		DBP	/	N.D.	Comply
		DIBP	/	N.D.	Comply

Remark:

(1) XRF result is on total Br while test item on restricted substances is PBBs/PBDEs. XRF result is on total Cr while test item on restricted substances is Cr(Ⅵ).

(2) The XRF result is for primary screening, and further chemical testing by ICP-OES (for Cd, Pb, Hg), UV-Vis (for Cr(Ⅵ)) and GC/MS (for PBBs, PBDEs) is recommended to be performed, if the concentration exceeds the below warning value according to IEC 62321-3-1: 2013. (unit: mg/kg)

Element	Polymer	Metal	Composite Materials
Cd	BL ≤(70-3σ)< X <(130+3σ)≤OL	BL ≤(70-3σ)< X <(130+3σ)≤OL	LOD < X <(150+3σ)≤ OL
Pb	BL ≤(700-3σ)< X<(1300+3σ)≤ OL	BL ≤(700-3σ)< X <(1300+3σ)≤ OL	BL ≤(500-3σ)< X<(1500+3σ)≤ OL
Hg	BL ≤(700-3σ)< X <(1300+3σ)≤ OL	BL ≤(700-3σ)< X <(1300+3σ)≤ OL	BL ≤(500-3σ)< X <(1500+3σ)≤ OL
Br	BL ≤ (300-3σ)< X	/	BL ≤ (250-3σ)< X
Cr	BL ≤ (700-3σ)< X	BL ≤ (700-3σ)< X	BL ≤ (500-3σ)< X

(3) BL=Below limit ,IN = Inconclusive, OL=Over limit, LOD=Limit of Detection;

(4) 1mg/kg = 1ppm = 0.0001%;

(5) “/ ”= Not Conducted or Not Applicable;

(6) MDL = Method Detection Limit;

(7) “N.D.”= Not Detected (Below Method Detection Limit);

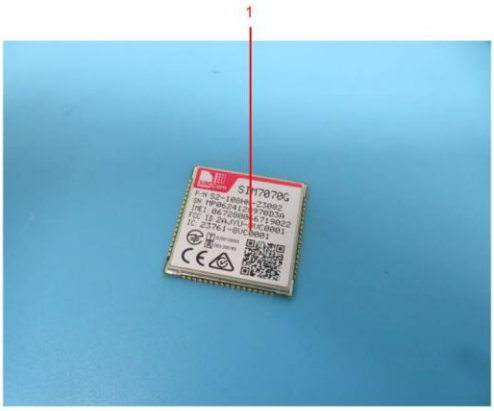
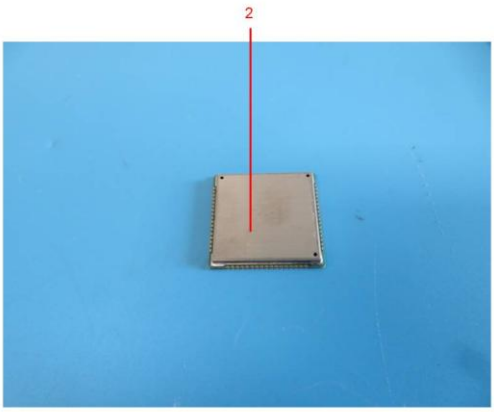
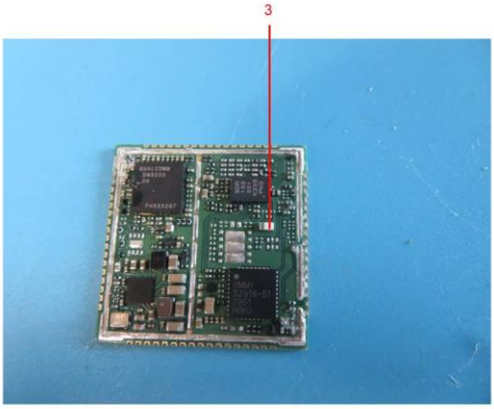
(8) For metal sample

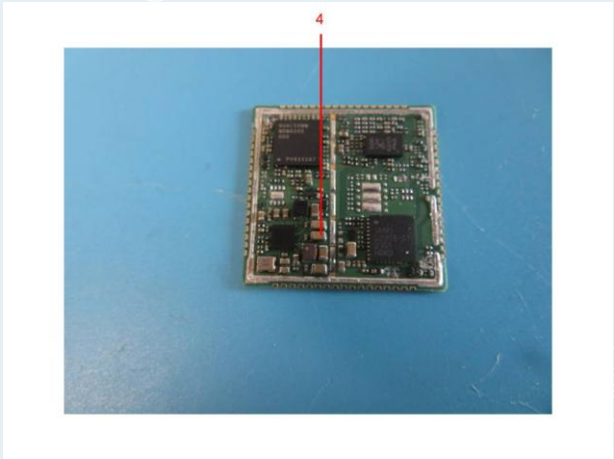
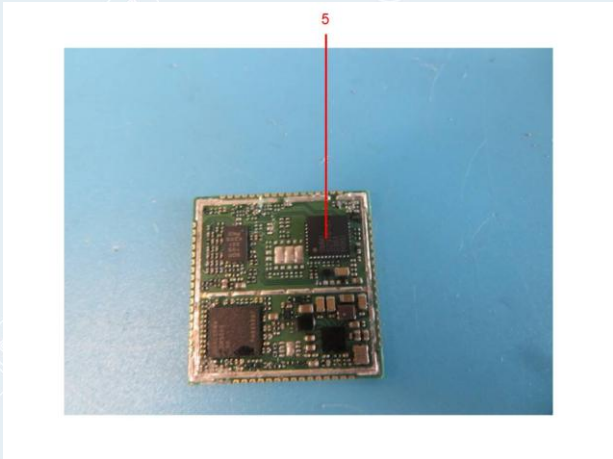
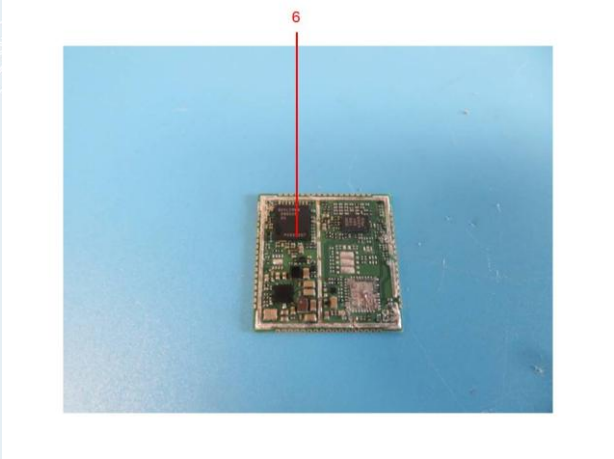
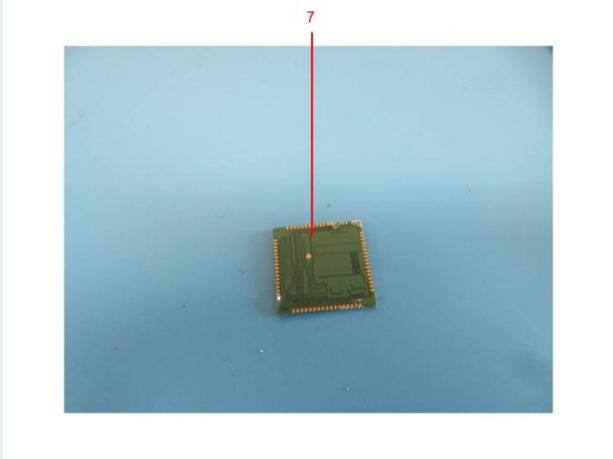
a. The sample is negative for Cr(VI) if Cr(VI) is N.D.(concentration less than 0.10 μg/cm^2).The sample

coating is considered a non- Cr(VI) based coating.

b. The sample is positive for Cr(VI) if the Cr(VI) concentration is greater than 0.13 μg/cm^2.The sample coating is considered to contain Cr(VI).

c. The result between 0.10 μg/cm^{2}and 0.13 μg/cm^2 is considered to be inconclusive. Unavoidable coating variations may influence the determination.

Photo of Sample	
	
	

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